



3500 ENGINE CYLINDER HEAD GROUP LIFTING TOOL

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

3500 ENGINE CYLINDER HEAD GROUP LIFTING TOOL

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December 17

4.03-160927-01

1.0 Introduction

Installing cylinder heads individually in 3500 engines (up to 16 in the 3516) can be a time consuming and labor intensive task, especially when considering the additional work of installing the intake and exhaust manifolds. Streamlining this process by using the lifting tool described in this best practice can improve installation times and improve the ergonomic work conditions for the technicians.

This Best Practice aims to facilitate and expedite the installation of the cylinder heads, intake and exhaust manifolds of the 3500 series engines by utilizing an easy to manufacture and use tool that allows the whole group of cylinder heads and manifolds to be installed as a single group.

2.0 Best Practice Description

The tool consists of a solid 1,5 x 1,5 square inch bar, 2,26m in length. There are 16 through holes 13.5mm in diameter, spaced 140mm apart along the longitudinal axis of the bar.

The bar has two rings of 3/8" ASTM A36 steel located symmetrically from its center where chains with hooks will attach to hoist the cylinder heads. The same tool is used for all 3500 series engines as the distance between fixing holes is the same.

3.0 Implementation Steps

- Review the status of the tool to verify the integrity of it before use.
- Check that the cylinder heads are properly assembled with your bank (Guides valves, springs, injectors, inlet and outlet valves, inserts, rotocoils) and manifolds installed (Fig. 1).
- Position lifting bar and attach the cylinder heads in each of the lower holes (Fig. 2 & 3)
- Lift assembly with cylinder heads & manifolds to the desired height (Fig. 4).
- Place the piece in its housing and remove the tools (Fig. 5).

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4.0 Benefits

The main benefits provided by this implementation are:

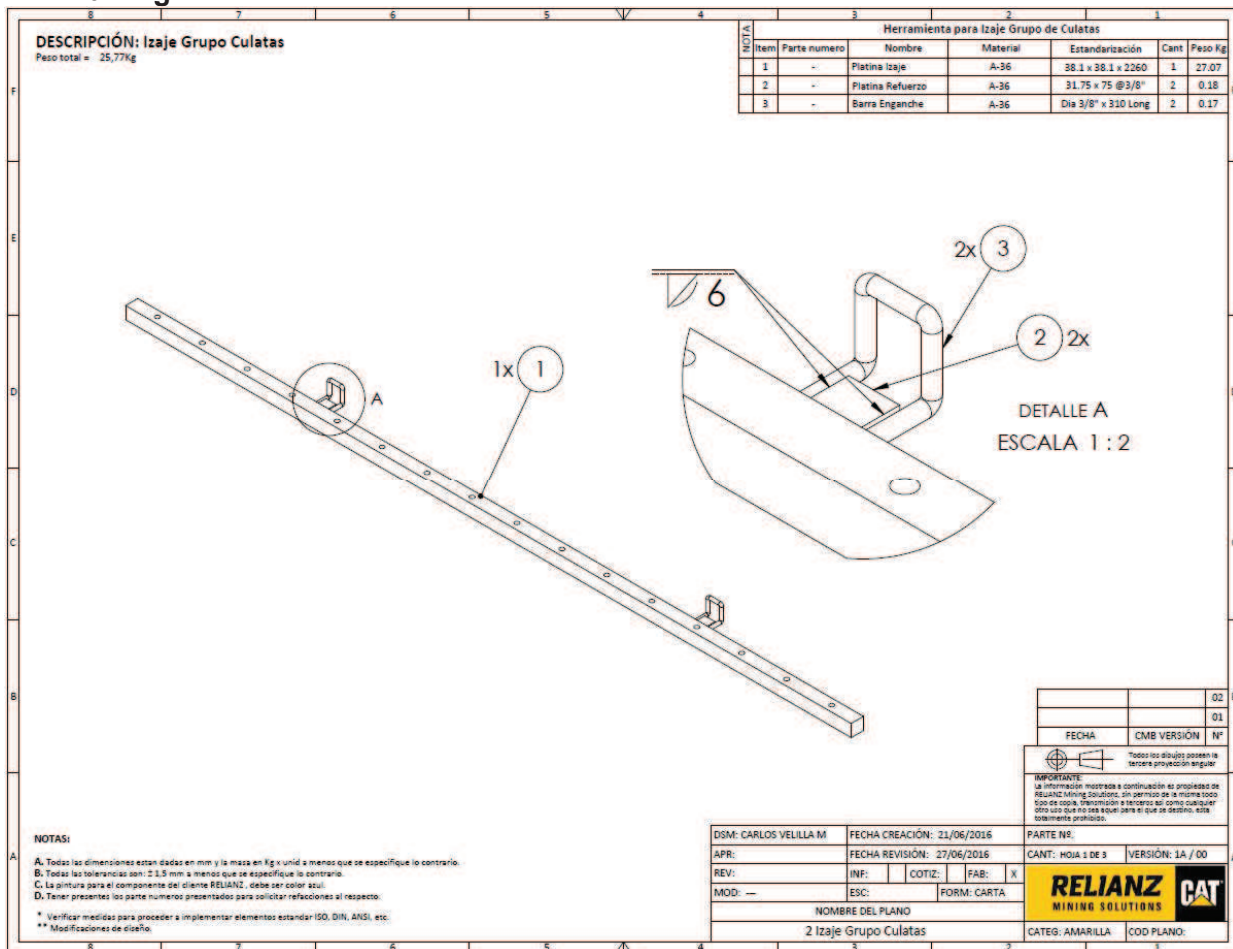
- Reduced installation time of the cylinder heads and manifold.
- Reduction of the movements necessary to install the cylinder heads.
- Improved ergonomic conditions for technicians.
- Easy to perform the work.
- Versatility, applies to arming and disarming the yokes 3500 series engines.

5.0 Resources Required

- Materials and equipment to manufacture the tool
- 1 welder

6.0 Supporting Attachments / References

- Drawing.



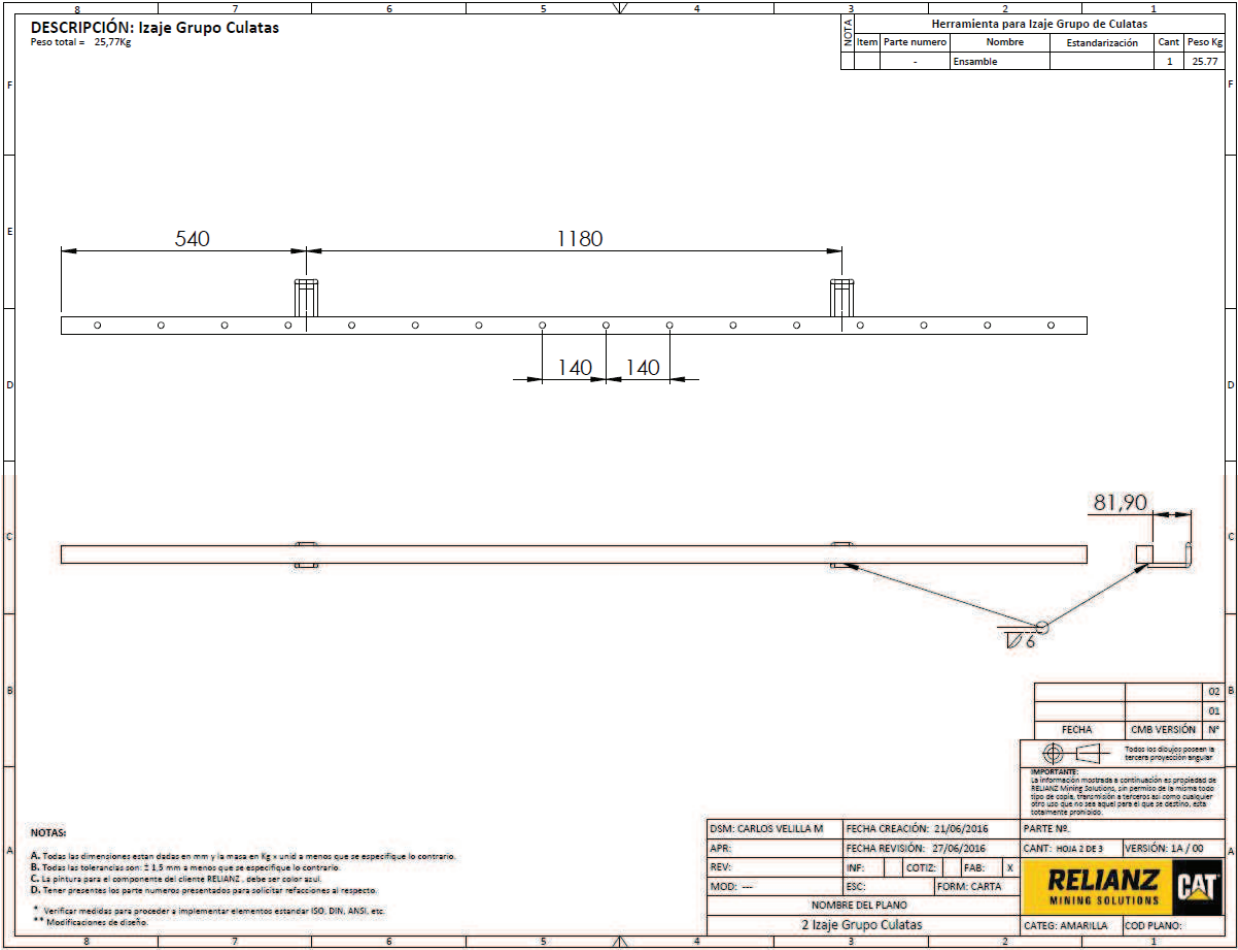
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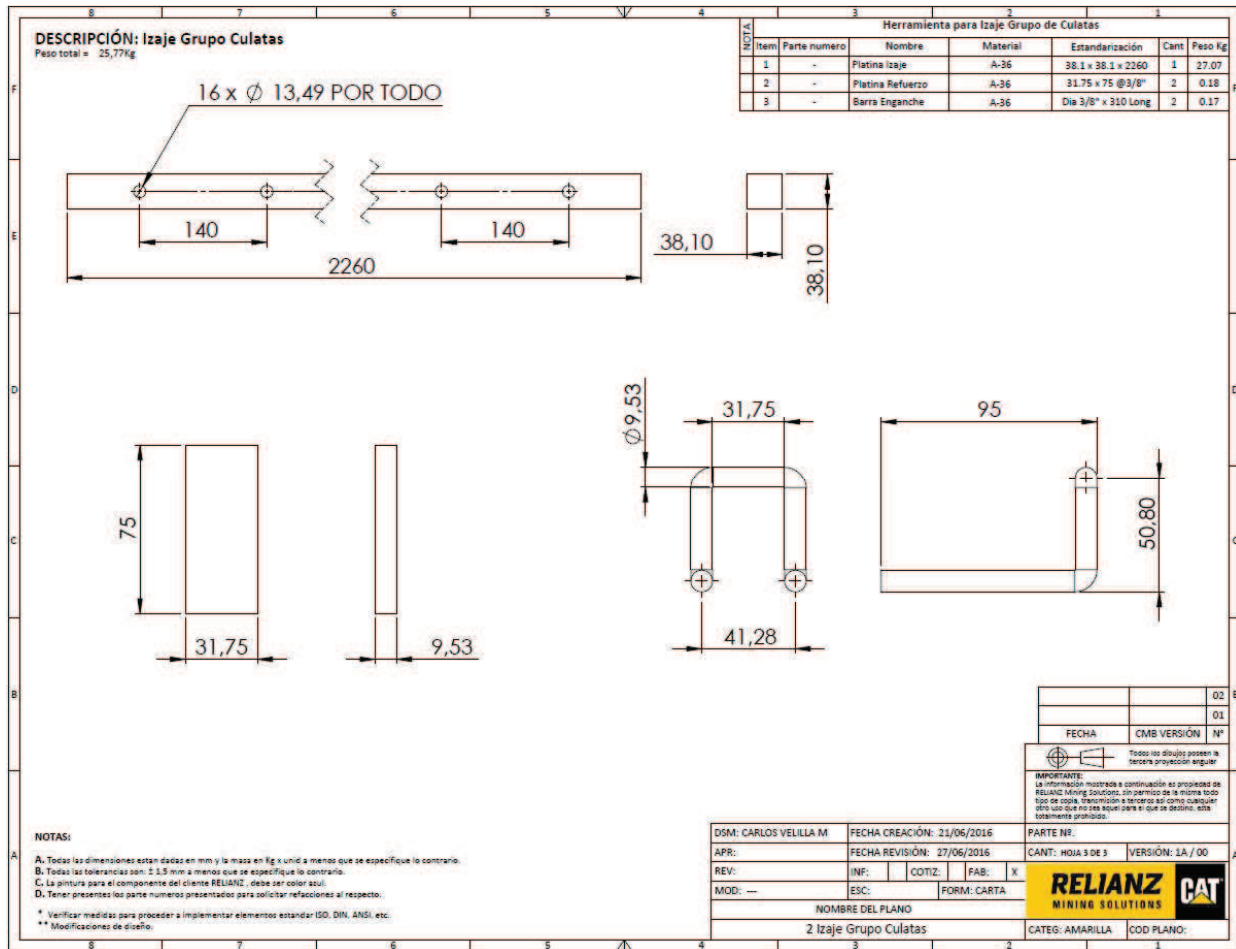
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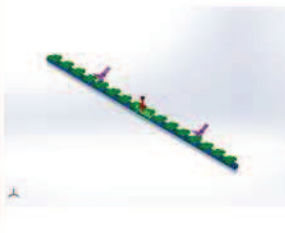
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• FEA: Finite Element Analysis

Propiedades de material

Referencia de modelo	Propiedades	Componentes
	Nombre: ASTM A36 Acero	Sólido 1(Combinar 1)(2 Izaje Grupo Culatas)
	Tipo de modelo: Isotrópico elástico lineal	
	Criterio de error predeterminado: Desconocido	
	Límite elástico: 2.5e+008 II/m^2	
	Límite de tracción: 4e+008 II/m^2	
	Módulo elástico: 2e+011 II/m^2	
	Coefficiente de Poisson: 0.26	
	Densidad: 7850 kg/m^3	
	Módulo cortante: 7.93e+010 II/m^2	
Datos de curva:II/A		

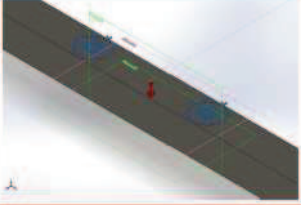
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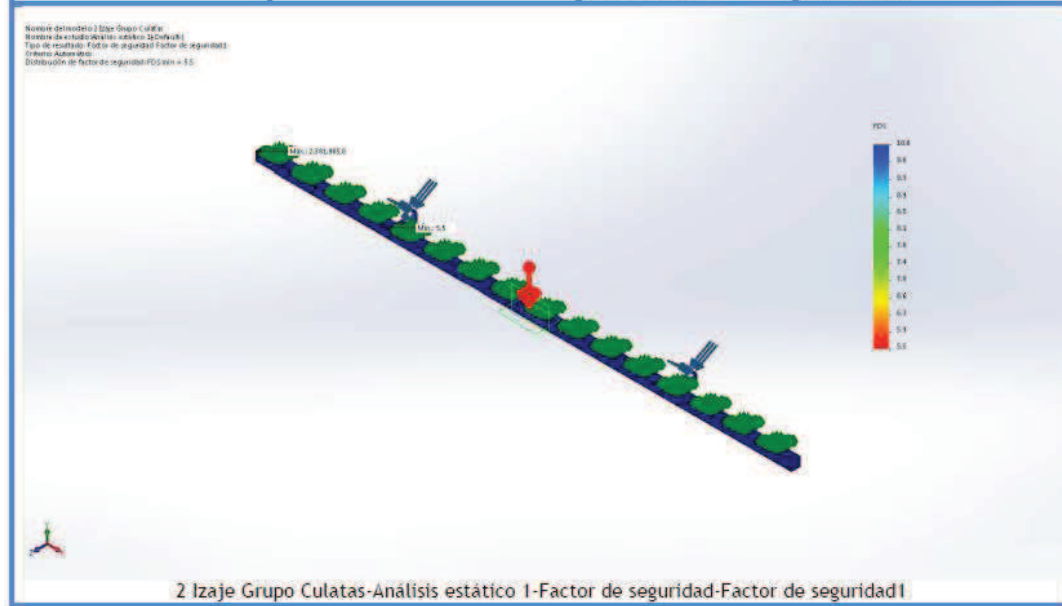
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Nombre de carga	Cargar imagen	Detalles de carga
Gravedad-1		Referencia: Planta Valores: 0 0 -9.81 Unidades: SI
Fuerza-1		Entidades: 2 cara(s) Tipo: Aplicar fuerza normal Valor: 353111

Nombre	Tipo	Min.	Máx.
Tensiones I	VOII: Tensión de von Mises	104.95411/m ² Ilo: 16311	4.5276e+00711/m ² Ilo: 24349

Nombre	Tipo	Min.	Máx.
Desplazamientos I	URES: Desplazamientos resultantes	0 mm Ilo: 349	0.169258 mm Ilo: 24418

Nombre	Tipo	Min.	Máx.
Factor de seguridad I	Automático	5.52169 Ilo: 24349	2.38199e+006 Ilo: 16311



- Maximum effort made in the tool: 45.2 Mpa, at the base of the ear lifting.
- Maximum strain exhibited: 0.17 mm

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- The minimum safety factor of the tool is 5.52. The concentration of maximum stress occurs at the point where the lifting lugs are attached, however, the safety factor indicates the tool is safe to use.

7.0 Related Best Practices

None

8.0 Acknowledgements

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